# SKILLPROOF

# Requirements Gathering

Stakeholder analysis, user stories, use cases, functional and non-functional requirements

| | |
|---|---|
| Program | DEPI |
| Track | React Front End |
| Instructor | Mohamed Amr |
| Team | SkillProof Team |
| Repository | github.com/nadaaradwano012-ui/SkillProof |
| Document Date | 15 October 2026 |

Document Status: Planning Baseline - subject to approved team and instructor updates.

# 1. Requirements Context

SkillProof is defined as a frontend-first React application for the DEPI React Front End track. React, JavaScript and CSS are confirmed. Authentication, backend, database, routing library, deployment and persistent storage are not finalized. The MVP may use clearly labeled mock data and mock role access while preserving replaceable data interfaces.

## 1.1 Product Goal

Help students understand how the skills in their profiles cover the required skills of selected internships, identify missing skills, plan learning actions and track applications. Support recruiters and company administrators with appropriately restricted application and internship views.

## 1.2 Core Business Rule

Skill Coverage = (Matched Required Skills / Total Required Skills) x 100

If no required skills are provided, the system returns no score. The score measures skill coverage only; it does not guarantee acceptance or prove mastery. Self-reported proficiency, assessment results, submitted evidence and reviewed evidence remain separate. Assessments are excluded from the first MVP.

# 2. Stakeholder Analysis

| Stakeholder | Needs | Value Provided | Key Restrictions |
|---|---|---|---|
| Student | Manage profile and skills; understand gaps; follow roadmap; track applications. | Clear skill coverage, missing skills and progress. | Cannot edit company data or hiring decisions. |
| Recruiter | Review applicants for authorized company internships and update statuses. | Organized candidate and application information. | Cannot edit student profile; sees only permitted data. |
| Company Admin | Create internships, define skills, manage recruiters and monitor hiring. | Central management of company opportunities and applicant stages. | Access limited to the company workspace. |
| Project Team | Clear ownership, stable interfaces and achievable scope. | Maintainable frontend and shared GitHub workflow. | Must not invent unapproved backend or integrations. |
| Instructor / DEPI | Evidence of planning, frontend learning, integration and documentation. | Traceable requirements and explainable implementation. | Claims must match actual implementation. |

## 2.1 Stakeholder Priorities

The first priority is a working student journey from profile and skills to internship requirements, matching results and roadmap. Company and recruiter views then complete the application workflow. Optional AI, assessment and enterprise verification features must not block the core MVP.

# 3. User Stories and Acceptance Criteria

| ID | As a | I want to | So that | Acceptance Summary |
|---|---|---|---|---|
| US-01 | Student | create an account and sign in | access the student workspace | Valid fields create a mock/frontend account flow; invalid input shows inline errors; role is read from the account, not selected during sign-in. |
| US-02 | Student | add, edit and delete a skill | keep the skill profile current | Empty names are rejected; duplicates ignore casing and surrounding spaces; delete requires confirmation. |
| US-03 | Student | add projects and certificates | support skill claims with links | Project/certificate fields validate required values and URLs; evidence is not automatically marked reviewed. |
| US-04 | Student | browse and save internships | find relevant opportunities | Search and filters work on available data; empty results are explained; saved state is visible. |
| US-05 | Student | run the Skill Gap Matcher | see matched and missing required skills | Input is validated; names are normalized; score follows the documented formula; no required skills returns no score. |
| US-06 | Student | add missing skills to a roadmap | turn gaps into learning actions | Selected missing skill creates a roadmap item without adding it to current skills. |
| US-07 | Student | update roadmap progress | track learning work | Status is one of Not Started, In Progress or Completed; overall completion uses completed items only. |
| US-08 | Student | track internship applications | know the current application stage | Statuses use the approved list; applications can be searched and filtered; details show a chronological timeline. |
| US-09 | Company Admin | create and publish an internship | make an opportunity visible to students | Drafts may be incomplete; publish requires title, work mode, deadline and at least one required skill. |
| US-10 | Company Admin | manage recruiters | control company access | Invitations show status; deactivation requires confirmation and preserves existing history. |
| US-11 | Recruiter | review an applicant | understand the application and skill coverage | Recruiter sees approved fields, Matched/Missing and evidence separately; score is informational only. |
| US-12 | Recruiter | update application status | move the candidate through hiring stages | Only approved statuses are available; update changes the timeline and is visible to authorized views. |
| US-13 | Company Admin | monitor hiring across internships | understand application progress | Dashboard summarizes real application counts without invented acceptance predictions or advanced analytics. |

# 4. Use Cases

## UC-01 Manage Student Skills

| Item | Definition |
|---|---|
| Primary Actor | Student |
| Preconditions | Student is in the student workspace. |
| Trigger | Student opens My Skills. |
| Main Flow | 1. View skills. 2. Add or edit a skill. 3. System validates and normalizes the name. 4. Save. 5. Updated list appears. |
| Alternative / Error | Empty or duplicate skill shows an inline error. Delete requires confirmation. |
| Postcondition | Valid skill data is available to the matcher. |

## UC-02 Run Skill Gap Matcher

| Item | Definition |
|---|---|
| Primary Actor | Student |
| Preconditions | Student skills and an internship with required skills are available. |
| Trigger | Student selects Run Analysis. |
| Main Flow | 1. Validate lists. 2. Normalize names. 3. Remove duplicates. 4. Compare each required skill. 5. Return Matched and Missing. 6. Calculate Skill Coverage. 7. Display explanation and disclaimer. |
| Alternative / Error | No required skills: no score. No student skills: all valid required skills are missing and the UI prompts the student to add skills. Invalid inputs show an error. |
| Postcondition | Result can be viewed and missing skills can be added to the roadmap. |

## UC-03 Manage Learning Roadmap

| Item | Definition |
|---|---|
| Primary Actor | Student |
| Preconditions | Student workspace is available. |
| Trigger | Student adds a missing skill or custom item. |
| Main Flow | 1. Create item. 2. Set priority and action. 3. Update status. 4. Recalculate completed-item progress. |
| Alternative / Error | Empty skill/action is rejected; duplicate handling follows the agreed roadmap rule. |
| Postcondition | Roadmap shows current priorities and progress. |

## UC-04 Create and Publish Internship

| Item | Definition |
|---|---|
| Primary Actor | Company Admin |
| Preconditions | Company admin workspace is available. |
| Trigger | Admin selects Create Internship. |
| Main Flow | 1. Enter details. 2. Add responsibilities. 3. Add required/preferred skills. 4. Assign recruiter optionally. 5. Save Draft or Publish. |
| Alternative / Error | Draft may be incomplete. Publish validates title, work mode, deadline and at least one required skill. Duplicates are rejected. |
| Postcondition | Published internship is visible in internship views; required skills can be matched. |

## UC-05 Submit and Track Application

| Item | Definition |
|---|---|
| Primary Actor | Student |
| Preconditions | Internship is open; student workspace is available. |
| Trigger | Student applies or logs an application. |
| Main Flow | 1. Create application with Applied status. 2. Add optional note. 3. View tracker. 4. Open details and timeline. |
| Alternative / Error | Duplicate application behavior follows the approved rule; invalid date or required fields show errors. |
| Postcondition | Application appears in student and authorized company/recruiter views. |

## UC-06 Review Applicant and Update Status

| Item | Definition |
|---|---|
| Primary Actor | Recruiter |
| Preconditions | Recruiter is authorized for the company internship. |
| Trigger | Recruiter opens an application. |
| Main Flow | 1. View permitted candidate fields. 2. Review Skill Coverage, Matched/Missing and evidence separately. 3. Add internal note. 4. Update status. |
| Alternative / Error | Unauthorized access is blocked; failed update preserves the current status and shows an error. |
| Postcondition | Authorized views show the updated application stage and timeline. |

## UC-07 Invite or Deactivate Recruiter

| Item | Definition |
|---|---|
| Primary Actor | Company Admin |
| Preconditions | Company admin workspace is available. |
| Trigger | Admin sends invitation or selects Deactivate. |
| Main Flow | Invitation: enter name/email and send. Deactivation: review consequences, select reassignment if available, confirm. |
| Alternative / Error | Expired invitation can be resent. If no replacement exists, assignments remain unassigned or deactivation is blocked according to final rule. |
| Postcondition | Recruiter access state and assignments are updated in the frontend data. |

# 5. Functional Requirements

| ID | Requirement |
|---|---|
| FR-01 | The system shall provide public Landing, Sign In and account-type pages. |
| FR-02 | The frontend shall support Student, Recruiter and Company Admin role views through mock role access until real authentication is approved. |
| FR-03 | The system shall redirect a signed-in user to the workspace associated with the stored role. |
| FR-04 | The student shall view and edit the approved profile fields. |
| FR-05 | The student shall add, edit, search, categorize and delete skills. |
| FR-06 | The system shall reject empty and normalized duplicate skill names. |
| FR-07 | The student shall manage projects, certificates and supporting links. |
| FR-08 | The system shall keep evidence status separate from skills and matching scores. |
| FR-09 | The student shall browse, search, filter, save and view internships. |
| FR-10 | The company admin shall create, edit, save as draft, publish and close internships. |
| FR-11 | The company admin shall define required and optional preferred skills. |
| FR-12 | Only required skills shall be included in the current Skill Coverage calculation. |
| FR-13 | The matching module shall validate, normalize and de-duplicate skill names. |
| FR-14 | The matching module shall classify required skills as Matched or Missing. |
| FR-15 | The system shall calculate Skill Coverage using the documented formula. |
| FR-16 | When no required skills exist, the system shall return no score and a clear message. |
| FR-17 | The matching logic shall remain independent from the React presentation components. |
| FR-18 | The student shall add missing skills or custom actions to the learning roadmap. |
| FR-19 | The roadmap shall support High, Medium and Low priority. |
| FR-20 | The roadmap shall support Not Started, In Progress and Completed status. |
| FR-21 | Roadmap completion percentage shall count completed items only. |
| FR-22 | The student shall add and view applications and filter them by approved status. |
| FR-23 | Application details shall show a chronological status timeline. |
| FR-24 | The recruiter shall view only applicants and internships permitted for the recruiter company/assignment. |
| FR-25 | The recruiter shall view permitted profile fields, Skill Coverage, Matched/Missing and evidence separately. |
| FR-26 | The recruiter shall add private internal notes visible only to authorized recruiter/company roles. |
| FR-27 | The recruiter shall update an application to an approved status. |
| FR-28 | The company admin shall view company-wide internships, applicants and hiring stages. |
| FR-29 | The company admin shall invite, view and deactivate recruiter accounts in the frontend flow. |
| FR-30 | Recruiter deactivation shall preserve existing application history and notes. |
| FR-31 | Destructive actions shall require confirmation. |
| FR-32 | Relevant pages shall provide loading, empty, validation, success and error states. |
| FR-33 | Displayed scores and counts shall be derived from current application data, not fixed design values. |

| ID | Requirement |
| --- | --- |
| FR-34 | The system shall label mock data and shall not present it as permanent database storage. |
| FR-35 | Assessment features shall not be included in the first MVP. |

# 6. Non-Functional Requirements

| ID / Area | Requirement |
|---|---|
| NFR-01 Usability | Primary tasks shall use clear labels, visible validation and understandable empty/error states. |
| NFR-02 Responsiveness | All MVP pages shall remain usable at desktop, tablet and mobile widths without horizontal overflow. |
| NFR-03 Accessibility | Forms shall use visible labels; focus states and keyboard access shall be visible; status shall not rely on color alone. |
| NFR-04 Performance | With local/mock data available, primary page interactions should target completion within 2 seconds on agreed test devices. |
| NFR-05 Reliability | The final demo flow shall contain no known critical uncaught errors. |
| NFR-06 Maintainability | Use readable React components, simple data flow, consistent naming and minimal approved dependencies. |
| NFR-07 Reusability | Matching logic and shared UI elements shall be separated from page-specific presentation where practical. |
| NFR-08 Security Boundary | Frontend route hiding is not treated as real authorization. Future backend services must enforce identity and permissions. |
| NFR-09 Privacy | Recruiter and company views shall display only fields approved for the relevant application context. |
| NFR-10 Data Integrity | Skill names shall be trimmed and compared case-insensitively for current matching and duplicate checks. |
| NFR-11 Explainability | Every Skill Coverage result shall expose matched count, required count, formula and disclaimer. |
| NFR-12 Compatibility | The application should target current mainstream desktop and mobile browsers available to the team. |
| NFR-13 Documentation | README and project documents shall reflect actual implementation and clearly identify limitations. |
| NFR-14 Testability | Matching logic shall accept explicit inputs and return reusable result data independent of the UI. |
| NFR-15 Honest Reporting | Tests, persistence, authentication and integrations shall not be reported as working unless verified. |

# 7. Data and Business Rules

| Rule ID | Rule |
|---|---|
| BR-01 | Normalize skill names by trimming surrounding spaces and comparing in a consistent case. |
| BR-02 | Duplicate skills shall not increase matched or required counts. |
| BR-03 | Each distinct required skill is counted once in the current formula. |
| BR-04 | Preferred skills are displayed separately and do not affect the current score. |
| BR-05 | No required skills means no score, not zero percent. |
| BR-06 | Evidence does not change the current Skill Coverage score. |
| BR-07 | A missing skill added to the roadmap does not become a current student skill automatically. |
| BR-08 | Completed roadmap progress = completed items / total roadmap items x 100. |
| BR-09 | Application statuses are Applied, Under Review, Interview, Accepted and Rejected. |
| BR-10 | Only authorized company/recruiter views may update hiring status or see internal notes. |

# 8. Matching Acceptance Test Matrix

| Case | Student Skills | Required Skills | Expected Result |
|---|---|---|---|
| Full match | HTML, CSS | HTML, CSS | Matched 2; Missing 0; Score 100% |
| Partial match | HTML, CSS, JavaScript, Git | HTML, CSS, JavaScript, React, Git | Matched 4; Missing React; Score 80% |
| No match | Python | HTML, CSS | Matched 0; Missing 2; Score 0% |
| Empty student list | None | HTML, CSS | Matched 0; Missing 2; Score 0% |
| Empty required list | HTML | None | No score |
| Casing difference | react | React | Matched 1; Score 100% |
| Whitespace difference | Git | Git | Matched 1; Score 100% |
| Duplicates | HTML, html | HTML, HTML | Distinct normalized skill counted once |
| Invalid entries | Empty/null values | HTML | Invalid values handled; valid skill still processed according to contract |

# 9. Requirements Traceability Summary

| Objective | Related Requirements | Primary Owner |
|---|---|---|
| Manage student profile and evidence | FR-04 to FR-08 | Member 2 |
| Explain skill coverage and gaps | FR-12 to FR-17, BR-01 to BR-06 | Members 1 and 3 |
| Track learning progress | FR-18 to FR-21, BR-07 to BR-08 | Member 4 |
| Track applications | FR-22 to FR-23, BR-09 | Member 4 |
| Manage internships and recruiters | FR-10 to FR-11, FR-28 to FR-30 | Member 5 |
| Review candidates and hiring stages | FR-24 to FR-27, BR-10 | Member 5 |
| Integration, quality and documentation | FR-31 to FR-35; NFR-01 to NFR-15 | Member 1 with all members |

# 10. Assumptions, Constraints and Open Decisions

## 10.1 Assumptions

- The first demonstrable version may use mock data.
- The team develops primarily in React with JavaScript and CSS.
- The design is simplified to remain implementable within the frontend track and deadlines.
- Company and recruiter permissions are represented in the frontend but require future backend enforcement for real security.

## 10.2 Constraints

- Limited current React and advanced JavaScript experience.
- Five team members and fixed official deadlines.
- No confirmed production backend, database, authentication service or deployment target.
- Assessment features are excluded from the first MVP.

## 10.3 Open Decisions

- Final routing package and shared state approach after training and repository review.
- Whether any backend, persistent storage or real authentication will be added later.
- Final deployment target.
- Whether recruiter assignment is mandatory before internship publication.
- Whether Withdrawn becomes an additional application status after MVP review.

# 11. Requirements Approval Baseline

These requirements define the proposed SkillProof MVP for instructor and team review. Changes to the score formula, role permissions, shared data structures or MVP scope require explicit agreement and corresponding documentation updates.

| | |
|---|---|
| Prepared By | SkillProof Team |
| Team Lead | Nada Adnan Radwan |
| Instructor | Mohamed Amr |

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|---|---|
| Baseline Date | 15 October 2026 |
| Repository | https://github.com/nadaaradwano012-ui/SkillProof.git |